Programming Quantum Computers (Apps IV: Cryptography)

(**Subtrack of** Quantum Computing: An App-Oriented Approach)

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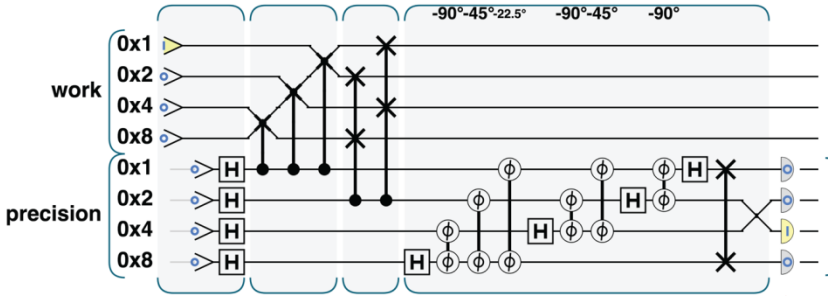
Sat., Jan. 4th, 2020

Quantum Computers are Real

- What are they <u>useful</u> for?
  - Let's discover, by programming them!

- A hands-on approach to programming QCs/QPUs.
  - By doing; i.e., by writing code & building programs.
  - Using simulators, since real QCs are harder-to-access (so far).

- Goals: Read, understand, write, and *debug* quantum programs.
  - Ones like this program.



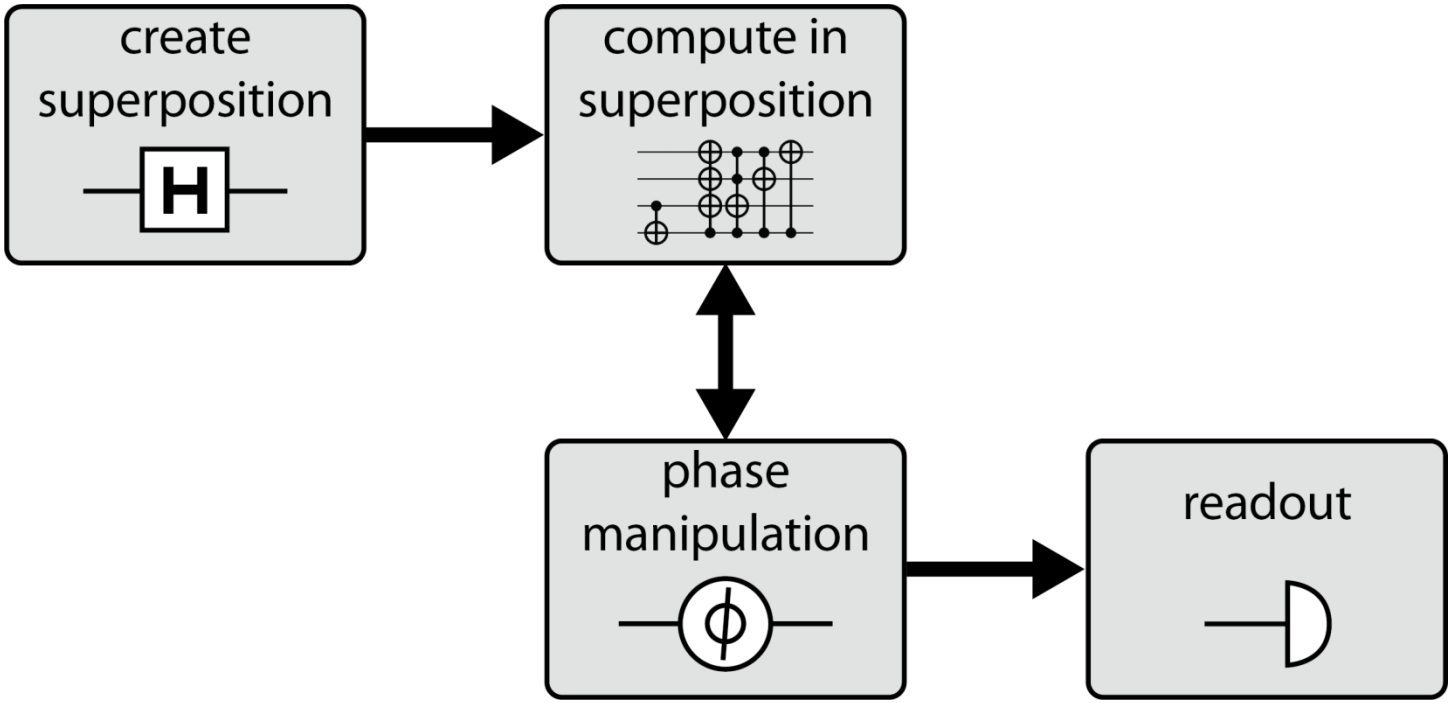
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Topics Covered

- Qubit, Superposition, Entanglement.
- Single-Qubit Ops: H, NOT and Phase.
- Multi-Qubit Ops: Conditional Ops.
- Teleportation.
- Quantum Arithmetic and Logic.
- (Quantum) Amplitude Amplification.
  - Converting phase info into magnitude info.
- Quantum Fourier Transform.
  - Revealing patterns (frequencies).
- (Quantum) Phase Estimation.
  - Characterization of quantum operations.
- QRAM, Vector & Matrix Quantum Encodings, and Quantum Simulation.

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Structure of Quantum Apps



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QUANTUM APPLICATIONS

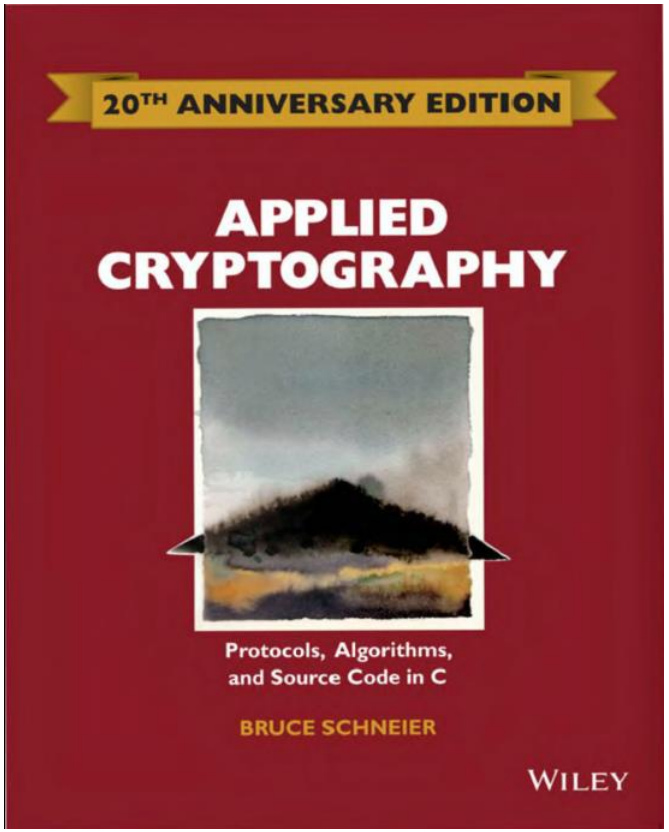
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QUANTUM CRYPTOGRAPHY (SHOR'S FACTORING ALGORITHM)

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Cryptography

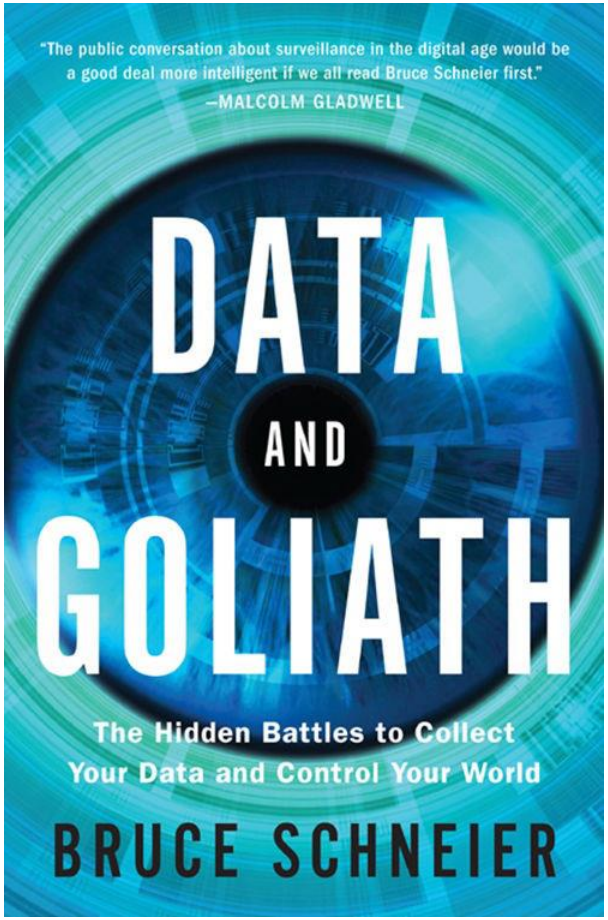
- Standard Reference (1995, 2015).
  - Discusses Shor's algorithm (§7.2), Quantum Key Distribution (QKD), and Quantum Cryptography (§23.16).
  - Starting point for wider interest in quantum computing.



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Data Security

- Schneier Strikes Again! (2015)
  - To the general non-academic populace.
  - Strongly recommended reading.
  - Nothing quantum.



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AC (1995): Assess at End of Lecture/ Studying Chapter

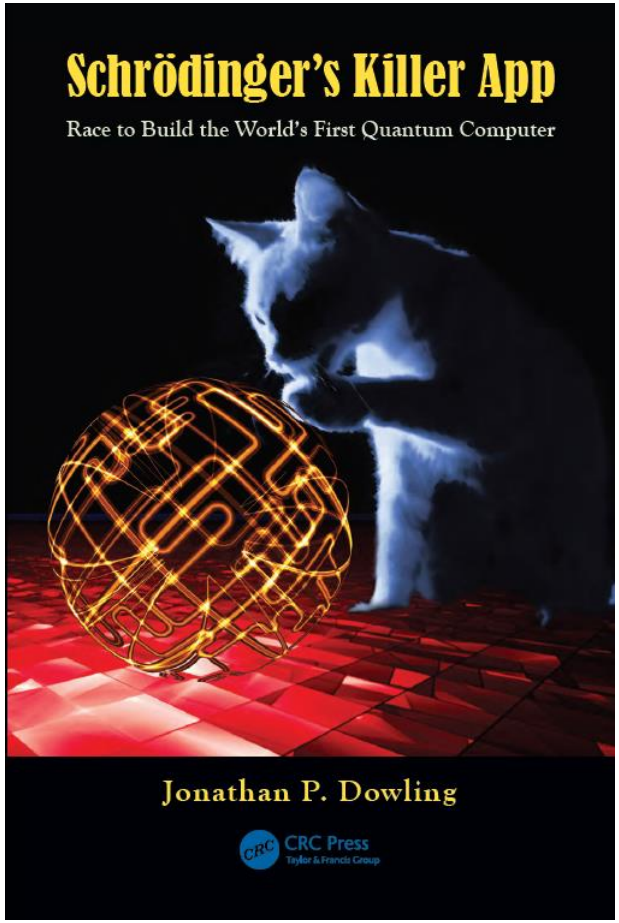
The consensus is that quantum computers are compatible with the fundamental laws of quantum mechanics. However, it is unlikely that a quantum factoring machine will be built in the foreseeable future . . . if ever. One major obstacle is the problem of decoherence, which causes superimposed waveforms to lose their distinctness and makes the computer fail. Decoherence will make a quantum computer running at 1° Kelvin fail after just one nanosecond. Additionally, an enormous number of gates would be required to build a quantum factoring device; this may render the machine impossible to build. Shor's design requires a complete modular exponentiator. No internal clock can be used, so millions or possibly billions of individual gates would be required to factor cryptographically significant numbers. If n quantum gates have some minimum probability p of failure, the average number of trials required per successful run is $(1/(1-p))^n$. The number of gates required presumably grows polynomially with the length (in bits) of the number, so the number of trials required would be superexponential with the length of the numbers used—worse than factoring by trial division!

So, while quantum factorization is an area of great academic excitement, it is extremely unlikely that it will be practical in the foreseeable future. But don't say I didn't warn you.

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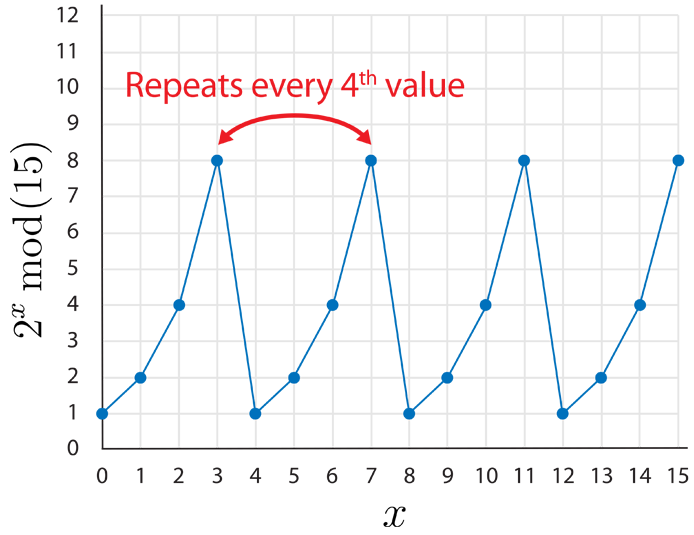
Quantum Cryptography

- The Quantum Killer App (Shor's algorithm and QKD).
  - Recommended reading.
  - The story of quantum cryptography and quantum computing.
  - Intelligence, government, and military agencies, worldwide, got interested in quantum computing.

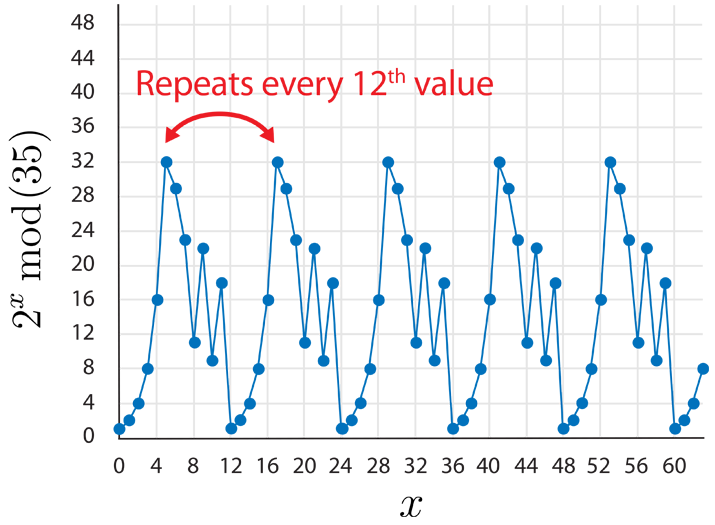


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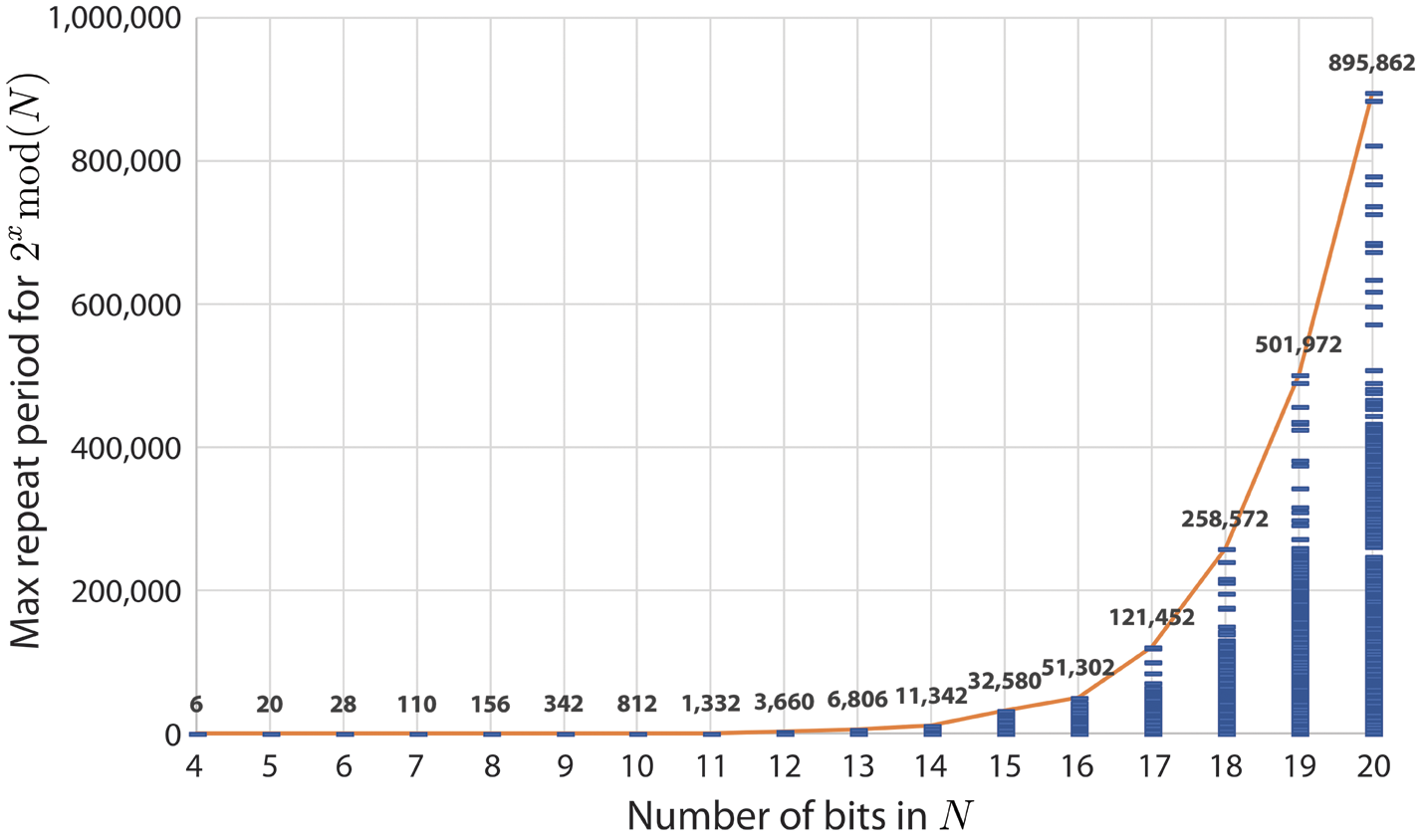
Factoring 15



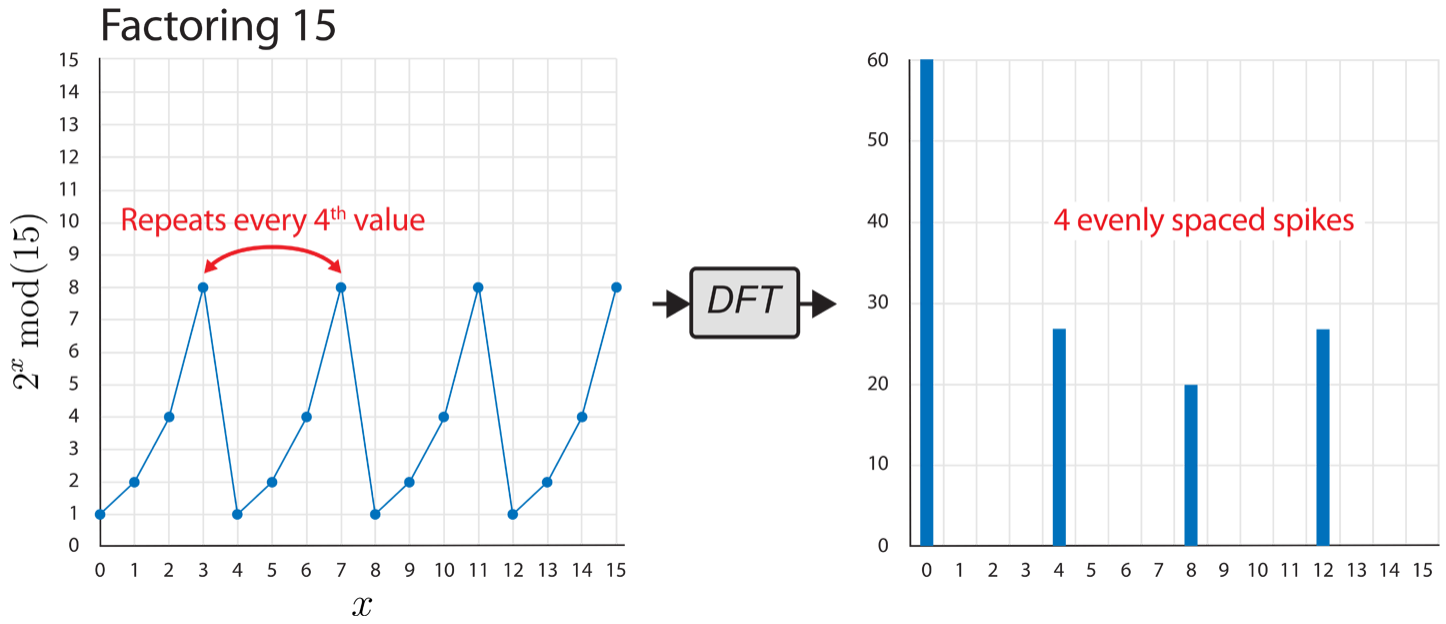
Factoring 35



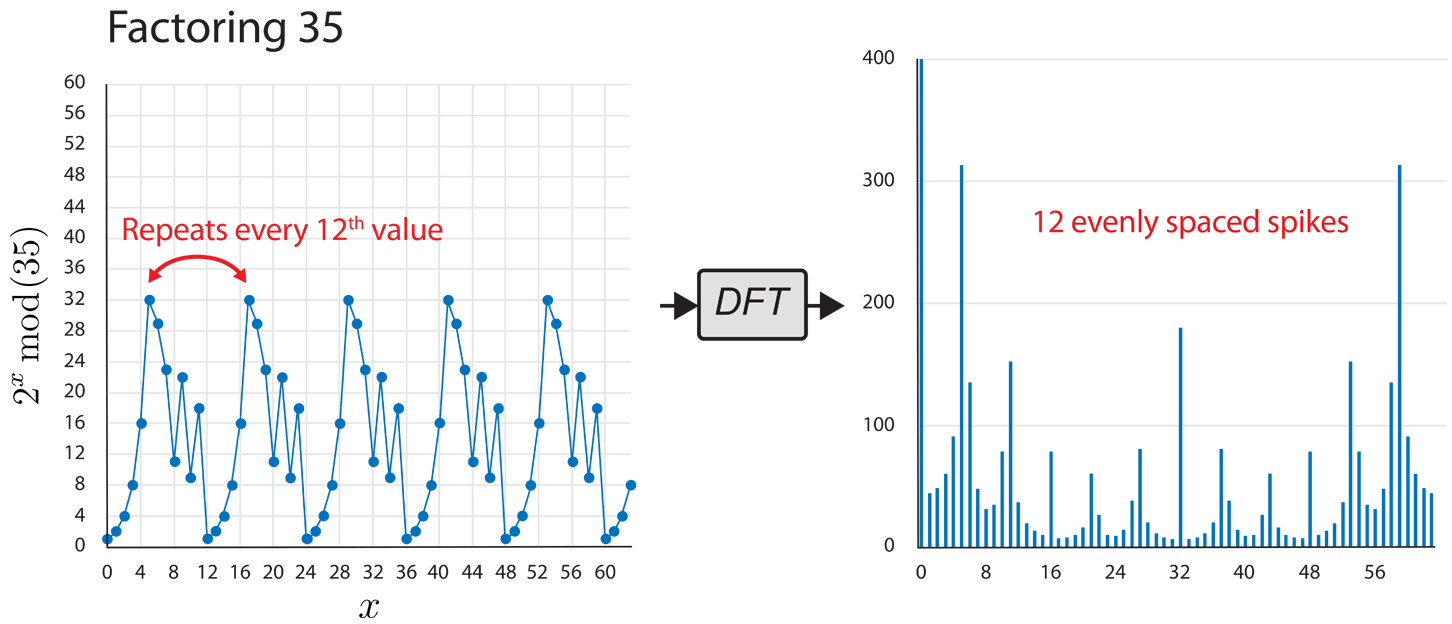
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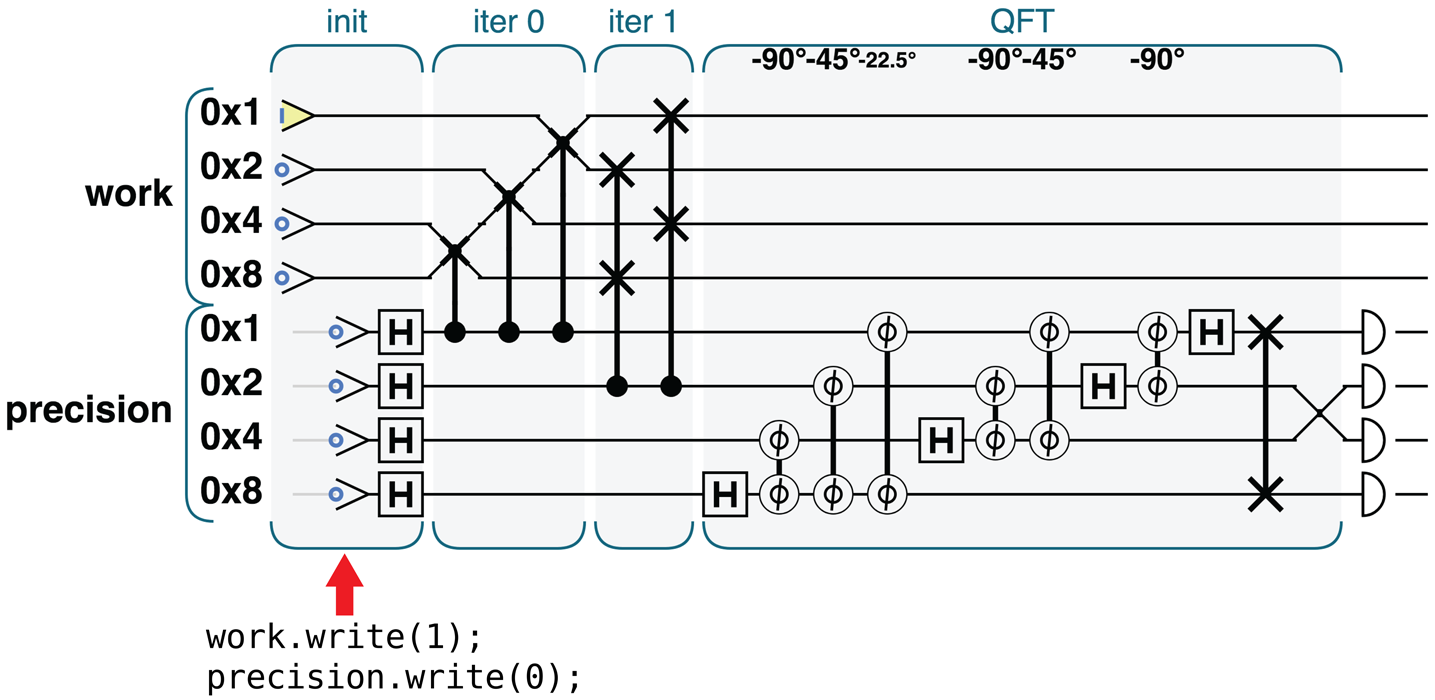


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Factoring 15: Step By Step

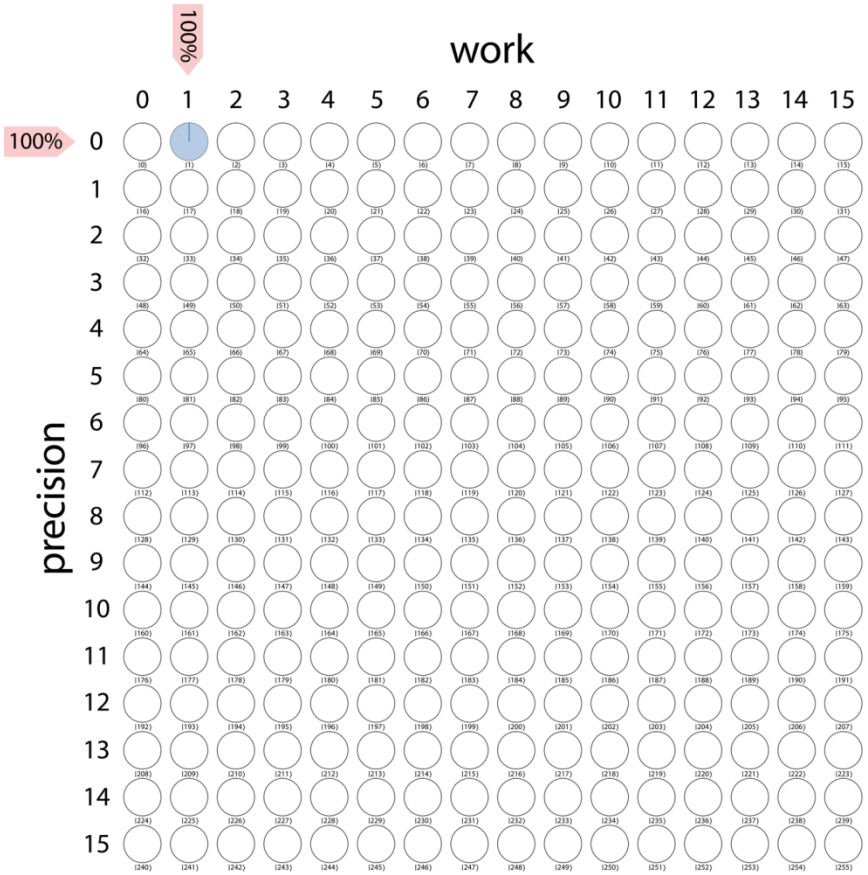
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Step 1: Initialize



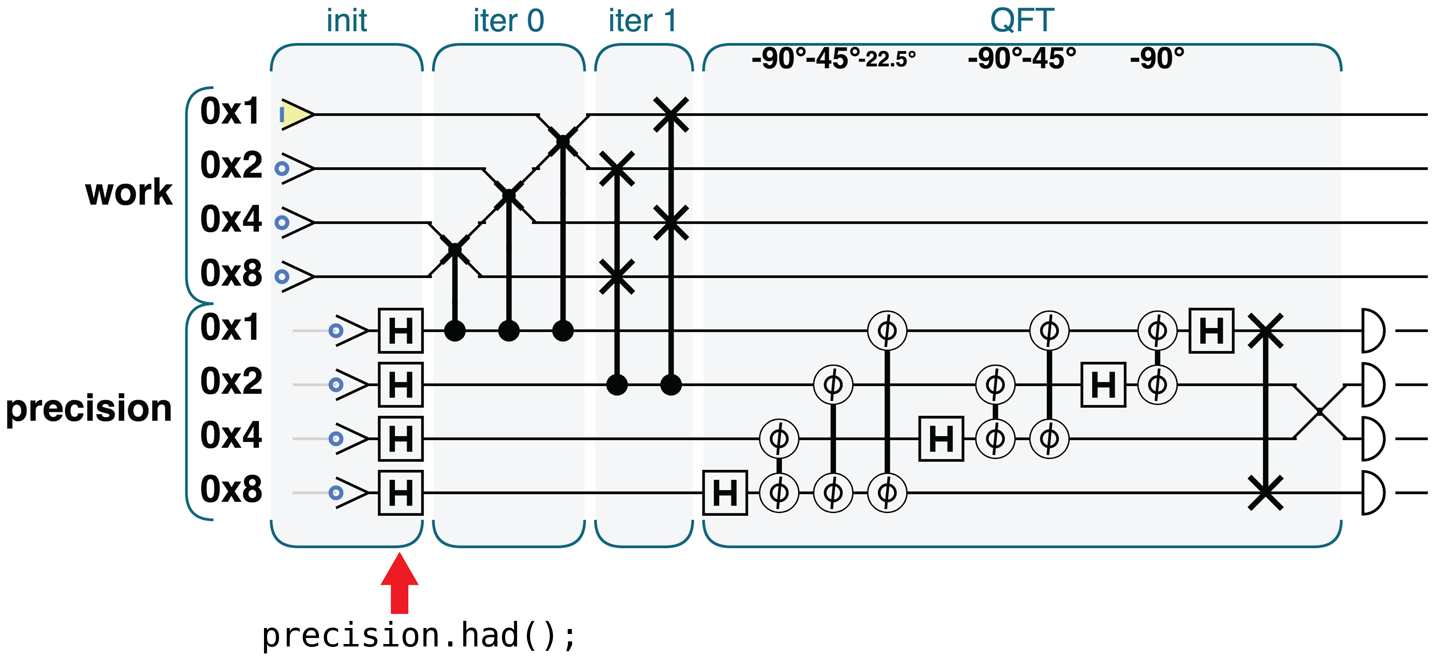
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Step 1: Initialize



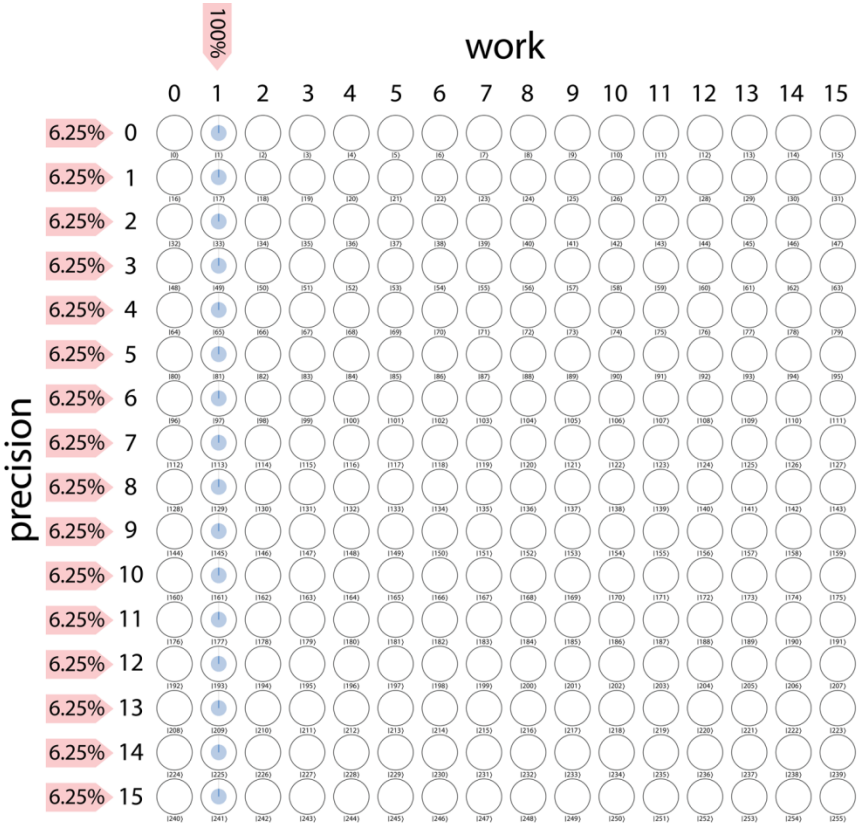
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Step 2: Expand Into Superposition



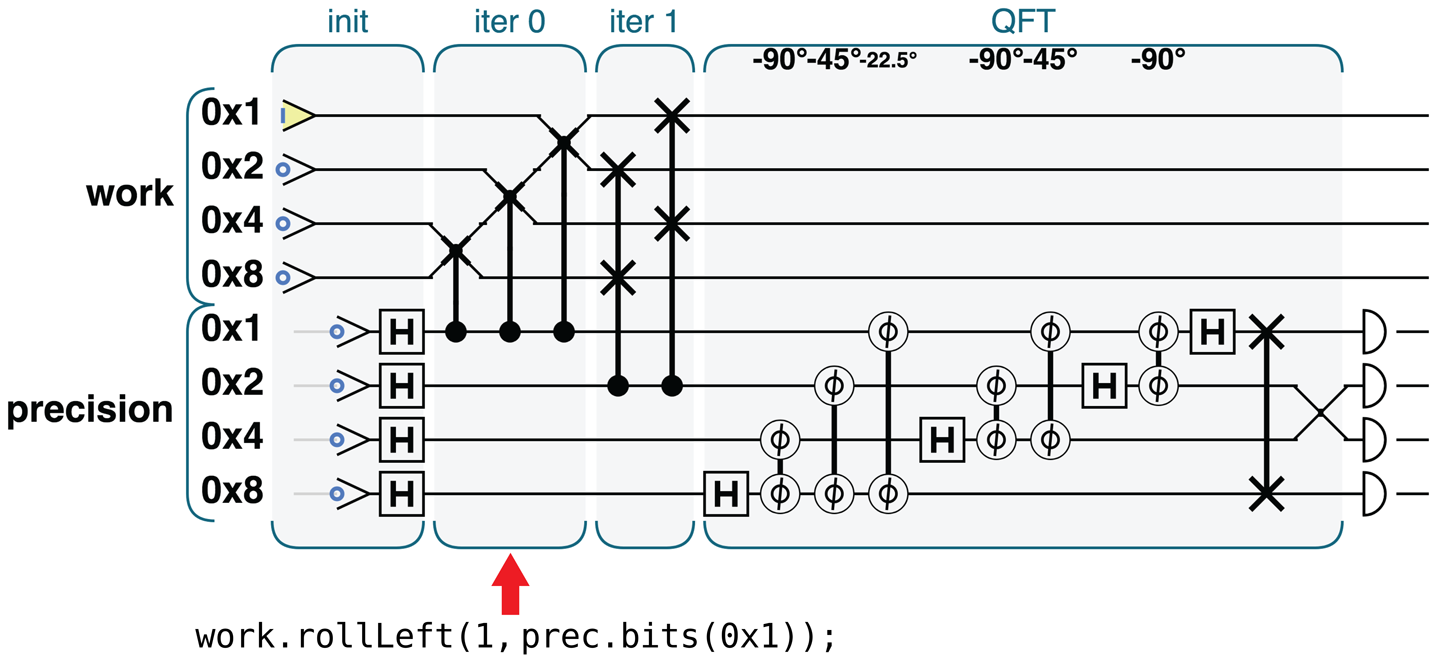
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Step 2: Expand Into Superposition



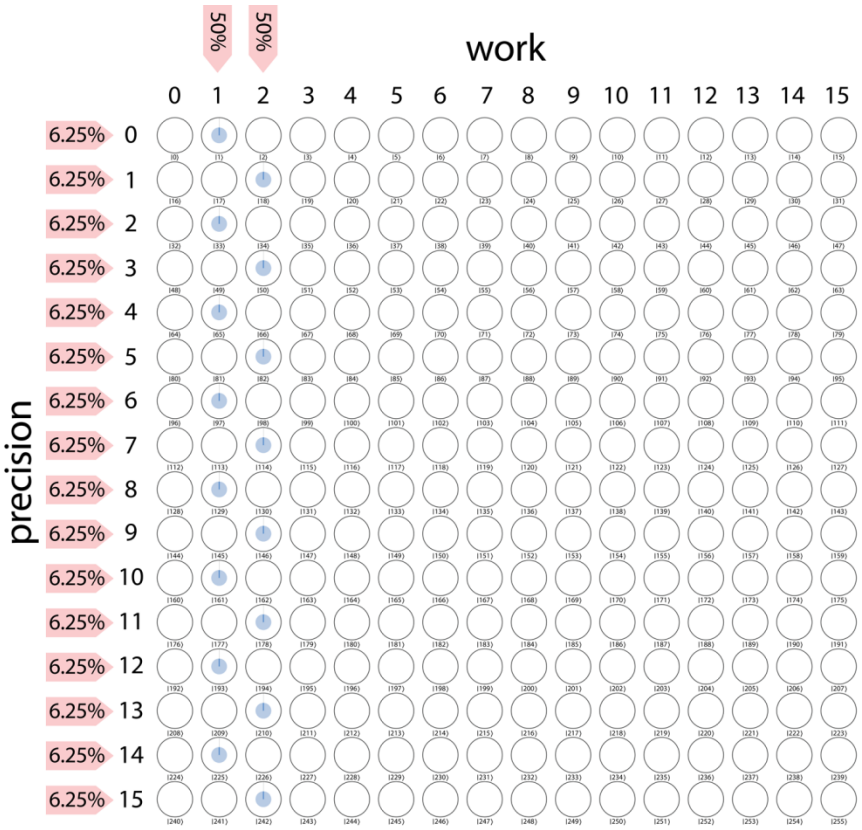
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Step 3: Conditional Multiply-by-2



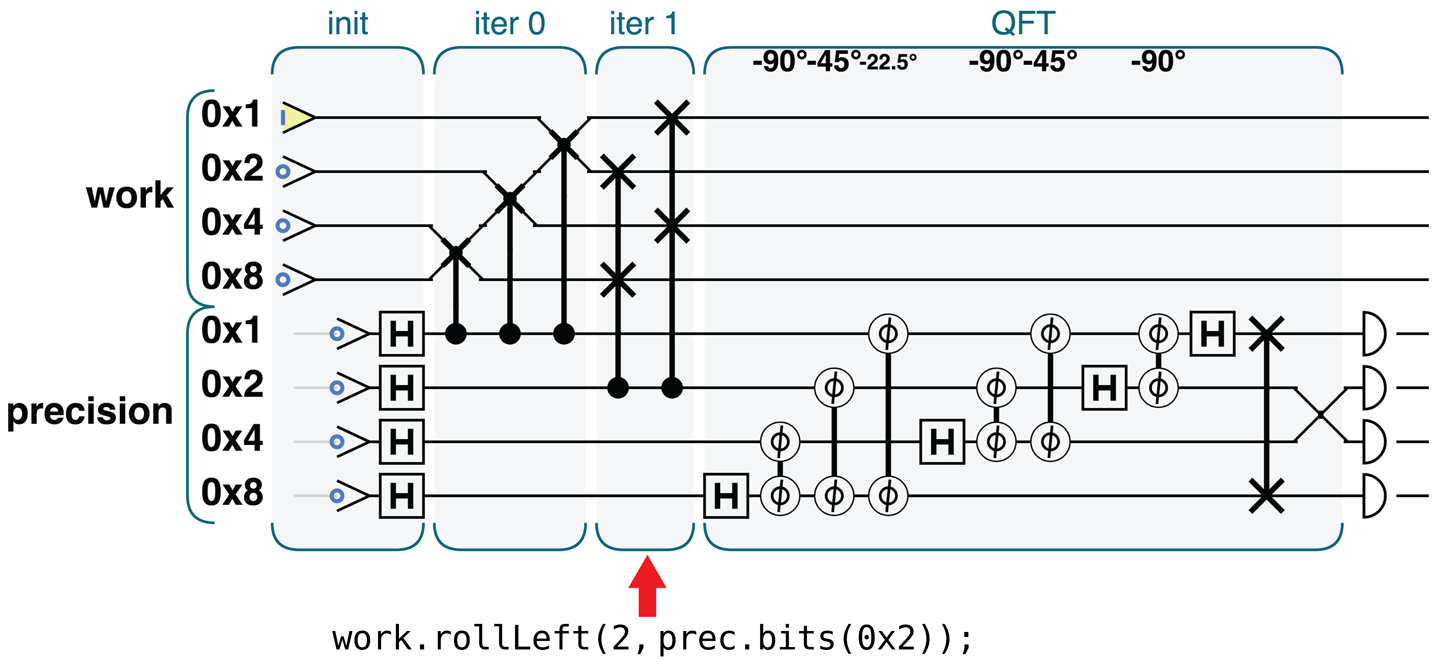
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Step 3: Conditional Multiply-by-2



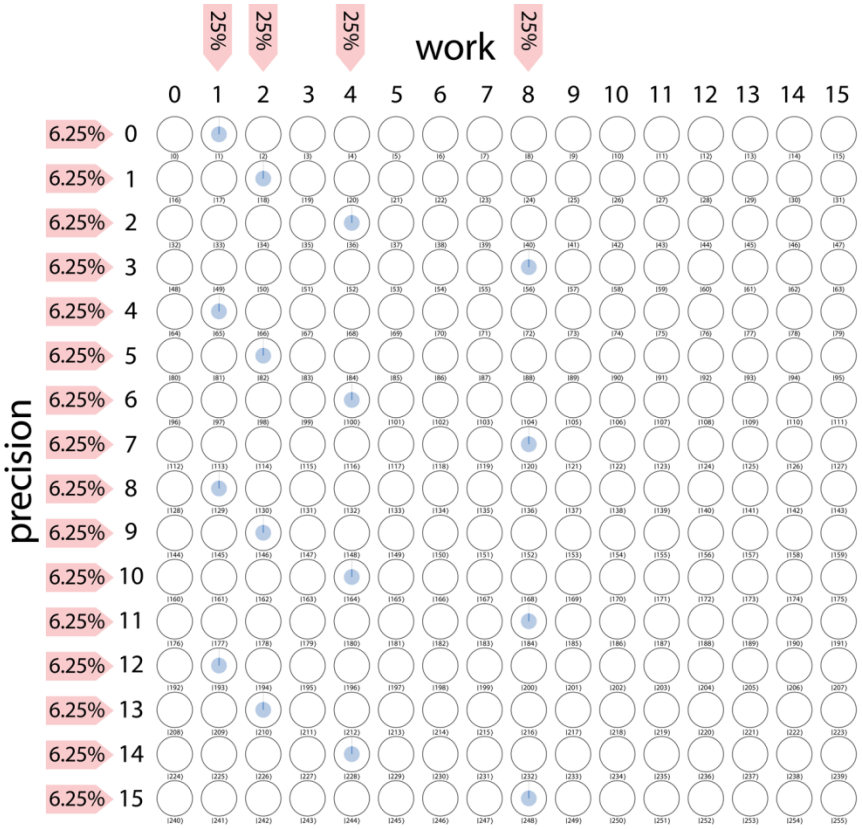
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Step 3: Conditional Multiply-by-4

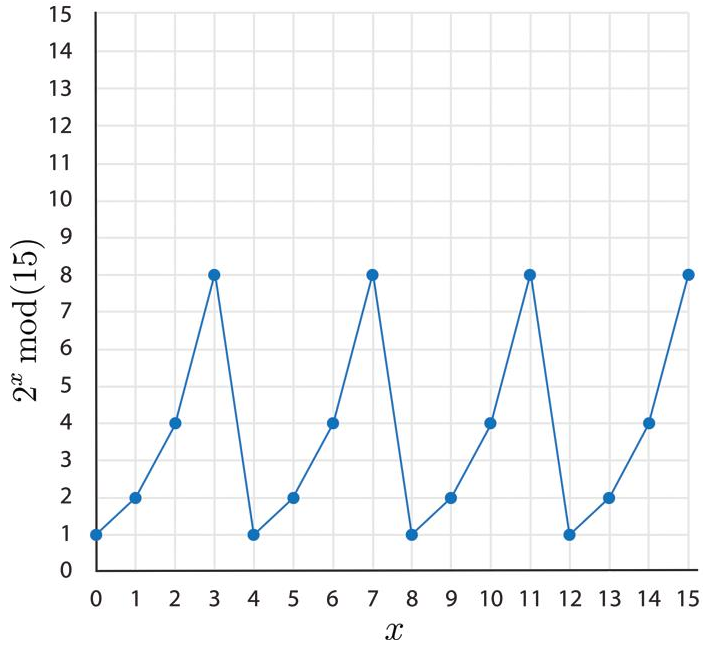


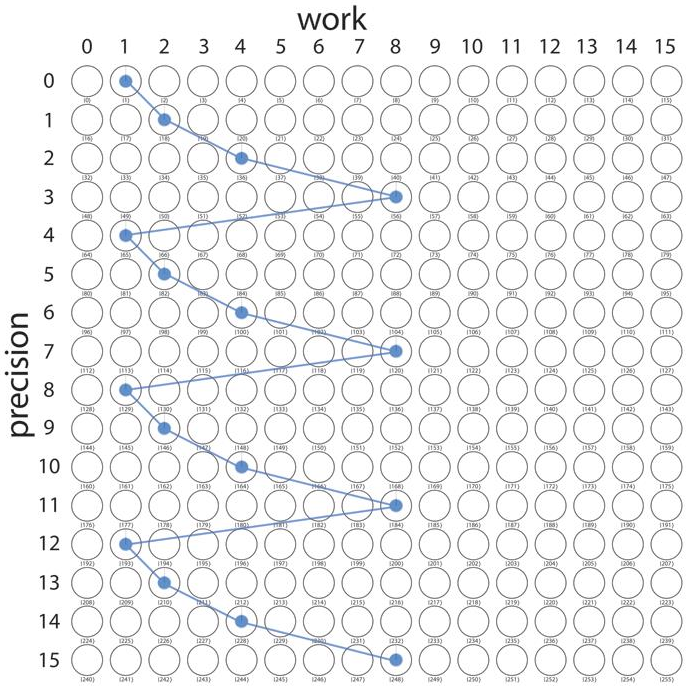
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Step 3: Conditional Multiply-by-4



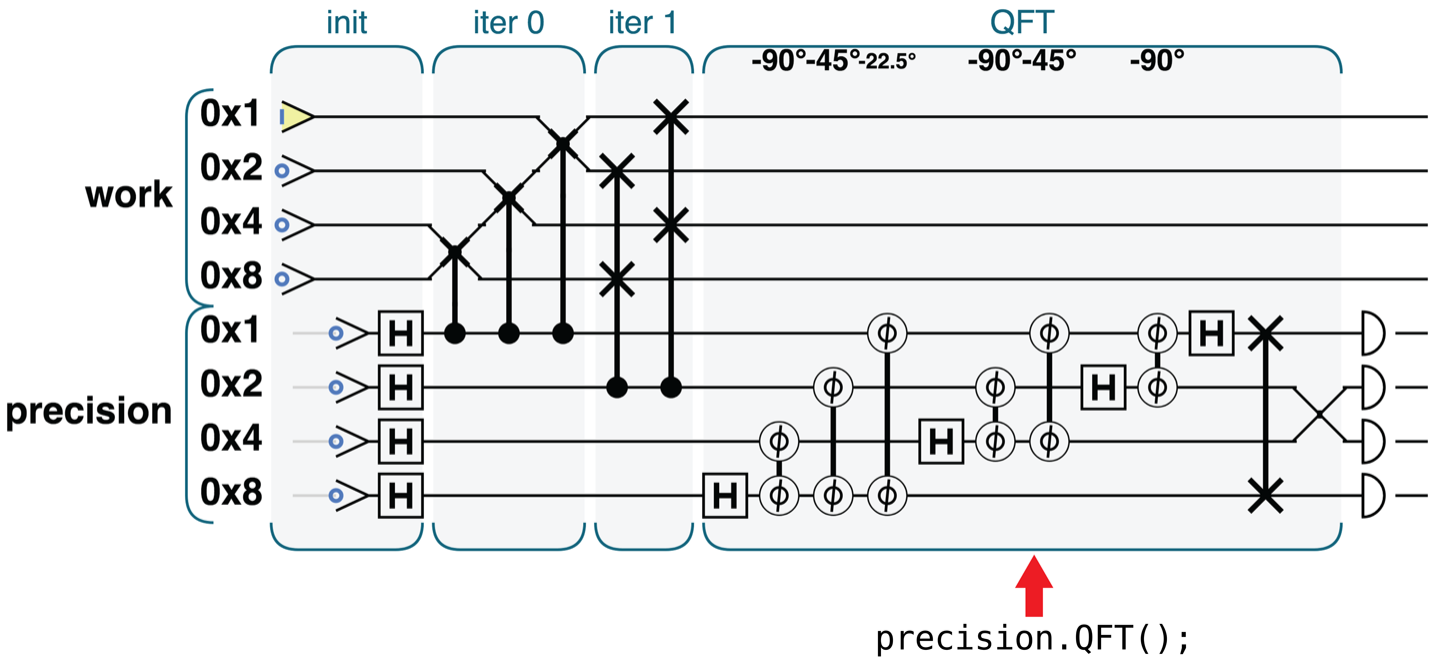
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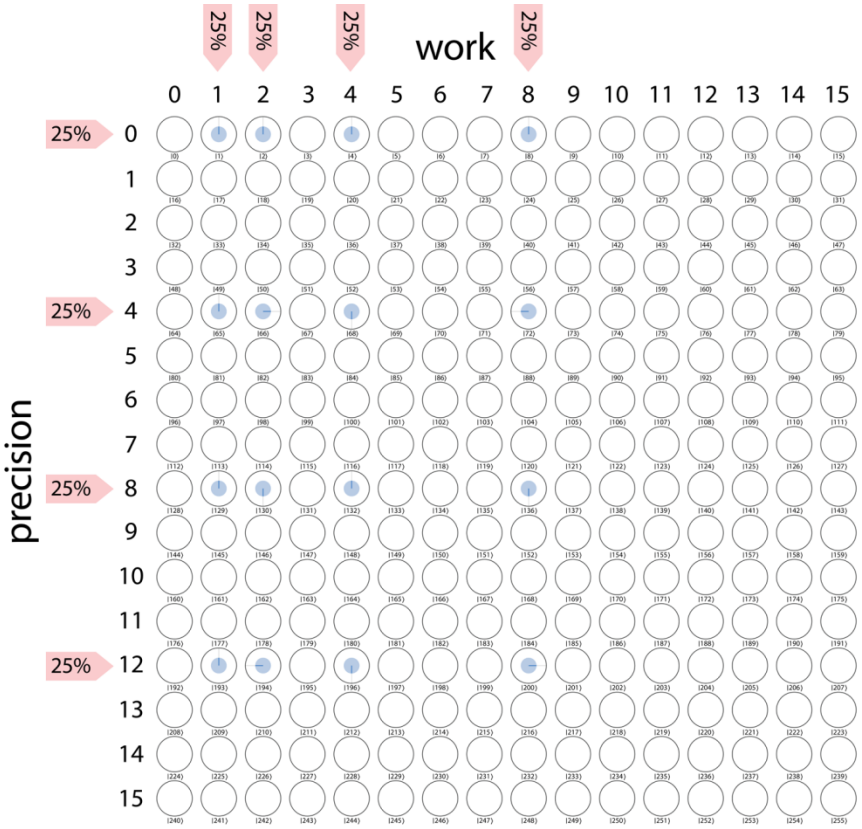
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Step 5: Quantum Fourier Transform



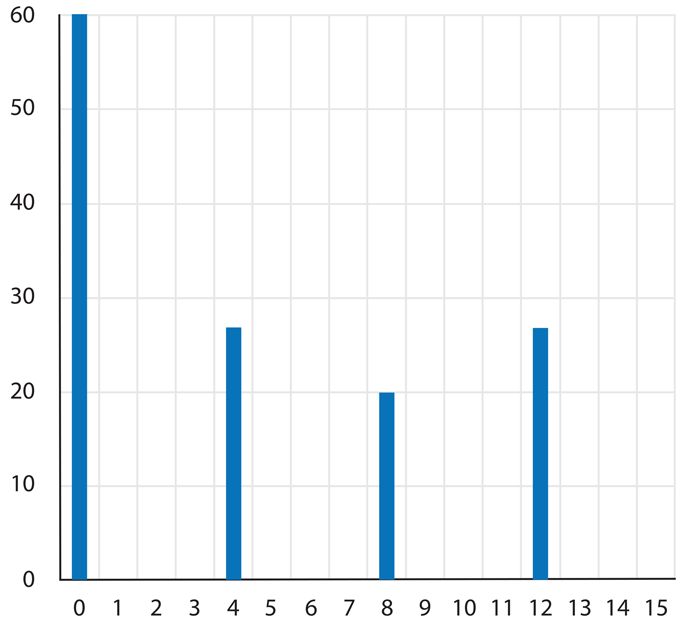
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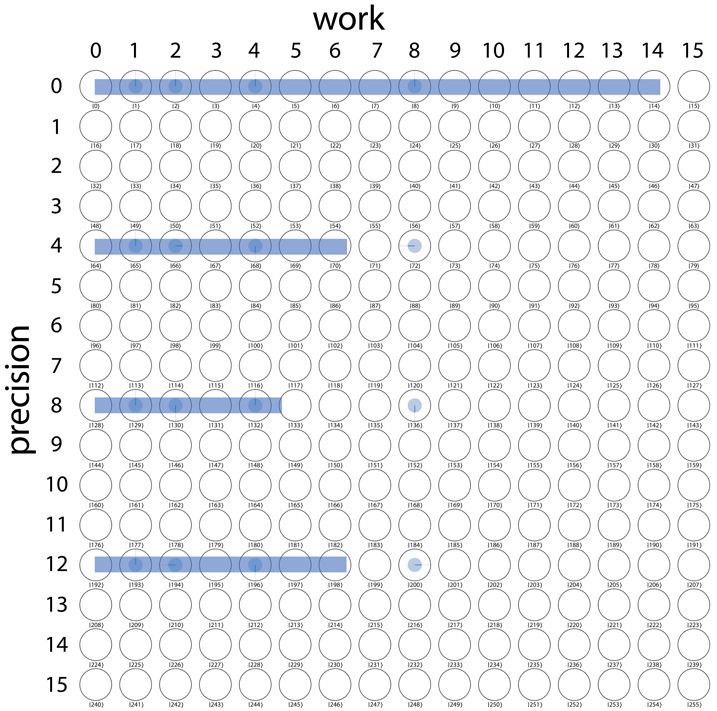
Step 5: Quantum Fourier Transform



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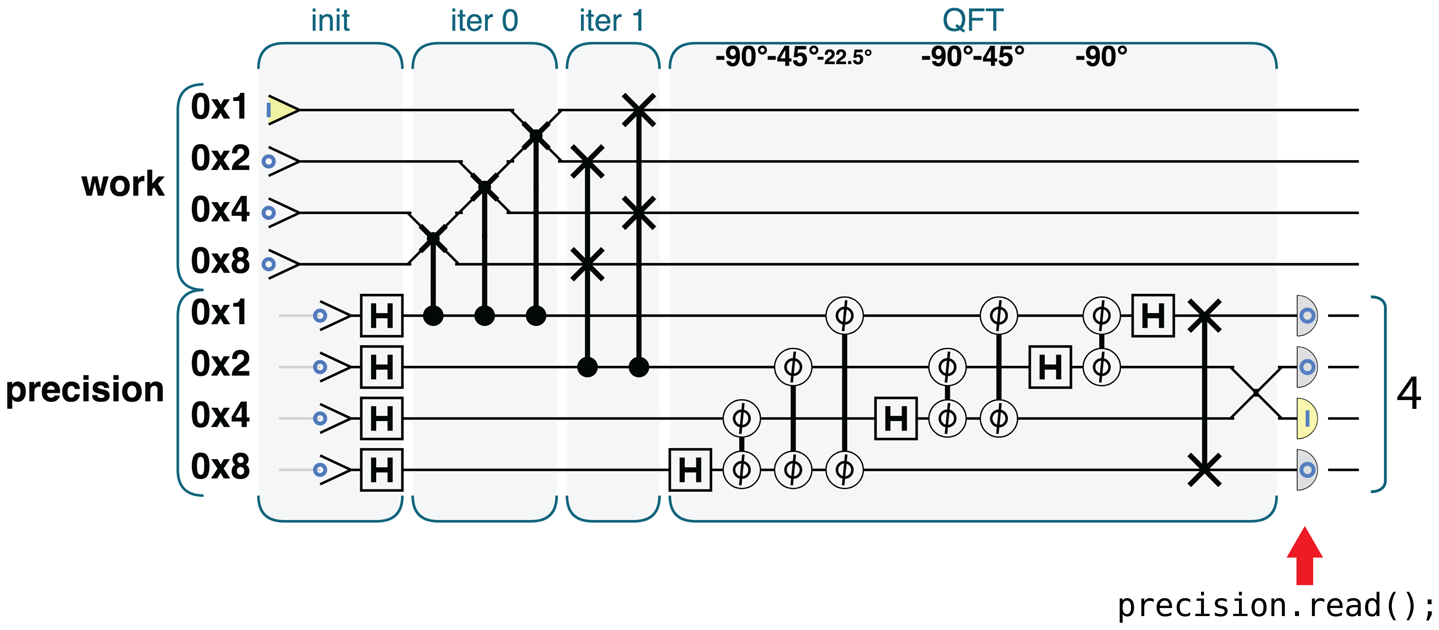
Step 5: Quantum Fourier Transform





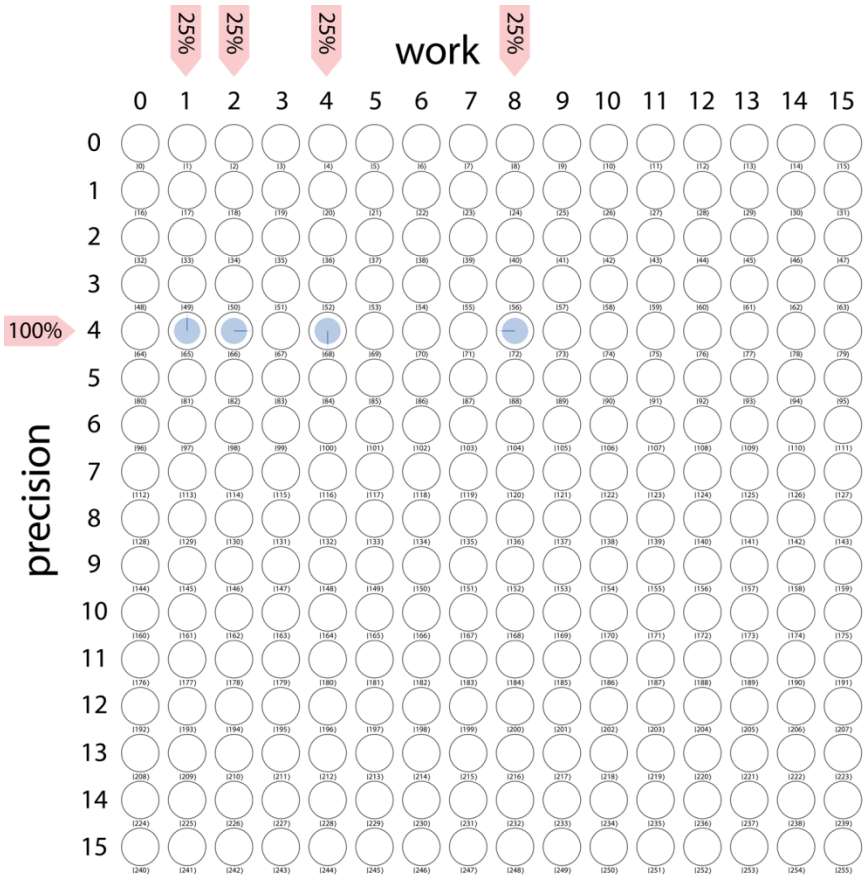
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Step 6: Read the Quantum Result



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Step 6: Read the Quantum Result



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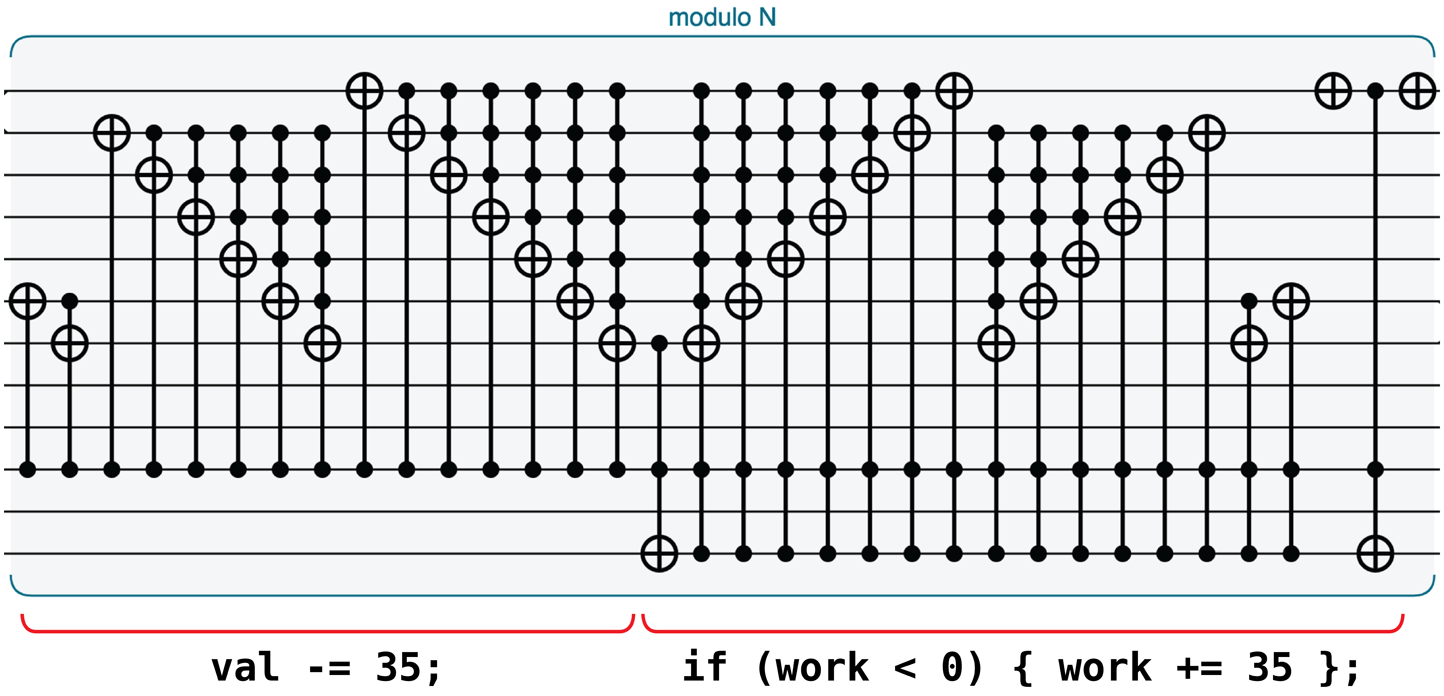
Step 7: Digital Logic

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Step 8: Check the Result

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Computing The Modulus



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Time Versus Space

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Coprimes Other Than 2

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Discussion

Q & A

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Next Lecture Appetizer!

- In next lecture (isA):
  - QML: Quantum Machine Learning
    - The two buzzwords!
    - HHL: Solving Linear Equations.
    - QPCA: The quantum SVD/KL/Hotelling transform.
    - QSVM.
    - Other QML apps.
  - Reading Ch.13 before next class is a MUST.

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Course Webpage

http://eng.staff.alexu.edu.eg/staff/moez/teaching/pqc-f19

- Where you can:
  - Download lecture slides (incl. exercises and homework).
  - Check links to other useful material.

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Thank You

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